

CY1010 – Environmental Chemistry

Practice Question Paper (50 marks)

Compiled from course slides — not an official department paper

Student name: _____

Roll No: _____

Department: _____

Each question carries 1 mark. Choose the single best answer and write the corresponding letter on the "Ans" line.

1. Which of the following best explains why Earth, unlike Mars and Venus, can support life?

- a) It has the strongest gravitational field in the Solar System
- b) It is the closest planet to the Sun
- c) It has liquid water, oxygen, and a moderate average temperature of about 15 °C
- d) It has the largest atmosphere by volume

Ans: _____

2. Approximately what percentage of Earth's total water is freshwater?

- a) 25%
- b) 10%
- c) 2.5%
- d) 50%

Ans: _____

3. Which of the following correctly gives the approximate composition of Earth's atmosphere by volume?

- a) N₂ 78%, O₂ 21%, Ar 0.93%, CO₂ ~0.04%
- b) O₂ 78%, N₂ 21%, CO₂ 0.93%, Ar 0.04%
- c) N₂ 50%, O₂ 50%
- d) CO₂ 78%, O₂ 21%, N₂ 0.93%

Ans: _____

4. Without the atmosphere, Earth's average surface temperature would be approximately

- a) +15 °C
- b) 0 °C
- c) -18 °C
- d) -63 °C

Ans: _____

5. Ocean water typically contains how much dissolved salt per litre, compared with drinking water?

- a) 33–37 g/L, versus less than 300 mg/L for drinking water
- b) 1–2 g/L, versus 300 mg/L for drinking water
- c) 100 g/L, versus 10 g/L for drinking water
- d) 300 mg/L, versus 33 g/L for drinking water

Ans: _____

6. The main components of LPG are

- a) Methane and ethane
- b) Propane and butane
- c) Butane and pentane
- d) Methane and propane

Ans: _____

7. Mercaptan is added to LPG and CNG mainly to

- a) Increase the calorific value

- b) Allow leak detection by smell
- c) Prevent corrosion of the cylinder
- d) Improve combustion efficiency

Ans: _____

8. CNG (compressed natural gas) is typically obtained at a pressure of about

- a) 2–5 MPa
- b) 20–25 MPa
- c) 220–2200 kPa
- d) 100 MPa

Ans: _____

9. Which coal type has the highest carbon content (about 90%) and is therefore classed as hard coal?

- a) Lignite
- b) Bituminous
- c) Anthracite
- d) Peat

Ans: _____

10. India holds roughly what share of the world's coal reserves, despite the coal being of relatively low heating value due to high ash content?

- a) 25%
- b) 15%
- c) 5%
- d) 40%

Ans: _____

11. Steam methane reforming (SMR), used to produce hydrogen, is represented by which reaction?

- a) $\text{CH}_4 + \text{H}_2\text{O} \rightarrow \text{CO} + 3\text{H}_2$
- b) $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$
- c) $3\text{C} + \text{O}_2 + \text{H}_2\text{O} \rightarrow \text{H}_2 + 3\text{CO}$
- d) $2\text{H}_2\text{O} \rightarrow 2\text{H}_2 + \text{O}_2$

Ans: _____

12. "Green hydrogen" refers to hydrogen produced by

- a) Steam methane reforming with CO_2 capture and storage
- b) Coal gasification
- c) Water electrolysis using renewable (solar/wind) electricity
- d) Steam methane reforming without CO_2 capture

Ans: _____

13. According to the slides, approximately what fraction of the world's hydrogen is currently produced by coal gasification?

- a) 50%
- b) 20%
- c) 4%
- d) 80%

Ans: _____

14. Given that global annual energy demand is about 500 EJ and Earth receives about 5.5 million EJ per year from the Sun, the slides estimate that humanity's annual energy demand is delivered by the Sun to Earth's surface in approximately

- a) One day
- b) One week
- c) One hour
- d) One month

Ans: _____

15. In the fission of U-235 shown in the slides ($^{235}\text{U} + \text{n} \rightarrow ^{236}\text{U} \rightarrow \dots + 3\text{n} + \text{energy}$), the two daughter nuclei produced are

- a) Ba-144 and Kr-89
- b) Sr-90 and Xe-140
- c) Cs-137 and Rb-96
- d) U-236 and He-4

Ans: _____

16. Why can uranium extracted from seawater be considered a renewable energy source?

- a) It is chemically converted into a non-radioactive form before use
- b) The extracted uranium is continuously replenished by steady-state reactions between seawater and rocks
- c) Seawater uranium does not undergo nuclear fission
- d) Reactors using seawater uranium require no fuel input at all

Ans: _____

17. In biodiesel production by transesterification, a triglyceride reacts with methanol (base catalyst) to give biodiesel (fatty acid methyl ester) and which byproduct?

- a) Glycerin
- b) Formaldehyde
- c) Ethylene glycol
- d) Acetone

Ans: _____

18. Biogas obtained by fermenting biomass typically contains methane and CO_2 in approximately what proportions?

- a) 90–95% methane, 5–10% CO_2
- b) 55–65% methane, 30–40% CO_2
- c) 20–30% methane, 60–70% CO_2
- d) 10% methane, 90% CO_2

Ans: _____

19. Second-generation biofuels are produced from which type of source material?

- a) Starch and sugarcane
- b) Waste lignocellulosic biomass such as sawdust, corncob, and bagasse
- c) Algae only
- d) Genetically engineered microorganisms only

Ans: _____

20. India's present blending levels for biodiesel and bioethanol, with 2030 targets in parentheses, are given in the slides as

- a) B2 (B5 by 2030) and E5 (E20 by 2030)
- b) B20 (B50 by 2030) and E10 (E30 by 2030)
- c) B5 (B10 by 2030) and E20 (E40 by 2030)
- d) B100 and E100 already achieved

Ans: _____

21. The catalyst conventionally used in the Haber–Bosch process for ammonia synthesis ($\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$) is

- a) Platinum
- b) Nickel
- c) Iron
- d) Palladium

Ans: _____

22. Ammonia synthesis is described in the slides as

- a) A negligible contributor to global energy consumption
- b) The single most energy-intensive industrial process, consuming about 1.5–2% of global energy

- c) A process requiring no catalyst
- d) Relevant only to explosives manufacturing

Ans: _____

23. In the Ostwald process, ammonia is first oxidised to NO ($4\text{NH}_3 + 5\text{O}_2 \rightarrow 4\text{NO} + 6\text{H}_2\text{O}$) using which catalyst?

- a) Pt catalyst
- b) Fe catalyst
- c) Ni catalyst
- d) V_2O_5 catalyst

Ans: _____

24. According to the slides, methane's atmospheric concentration has changed from a pre-industrial level to today's level of approximately

- a) 0.7 ppm to about 1.8 ppm
- b) 7 ppm to 18 ppm
- c) 270 ppb to 340 ppb
- d) 0.04% to 0.4%

Ans: _____

25. The main source of nitrous oxide (N_2O) in the atmosphere, as described in the slides, is

- a) Combustion of fossil fuels alone
- b) Denitrification of nitrate and nitrite in soil
- c) Photodissociation of N_2 in the stratosphere
- d) Volcanic outgassing

Ans: _____

26. The average atmospheric residence time of nitrous oxide given in the slides is about

- a) 2 weeks
- b) 12 years
- c) 120 years
- d) 3200 years

Ans: _____

27. Sulphate particulate matter in the atmosphere, which strongly absorbs IR radiation, is primarily formed by

- a) Direct emission from vehicle tailpipes
- b) Oxidation of sulfur dioxide (SO_2)
- c) Photolysis of CO_2
- d) Condensation of water vapour

Ans: _____

28. Compared with other greenhouse gases, carbon particulate matter emitted from diesel engines has an atmospheric residence time of about

- a) 2 weeks
- b) 12 years
- c) 120 years
- d) 3200 years

Ans: _____

29. Which fluorinated compound is identified in the slides as having the highest global warming potential, about 22,800 times that of CO_2 ?

- a) HFC-134a
- b) CFC-11
- c) Sulphur hexafluoride (SF_6)
- d) Tetrafluoromethane (CF_4)

Ans: _____

30. Perfluorocarbons (PFCs) such as CF_4 and C_2F_6 are mainly generated as byproducts of

- a) Ammonia synthesis
- b) Aluminium production and semiconductor manufacturing
- c) Biodiesel production
- d) Cement manufacturing

Ans: _____

31. Tropospheric ("bad") ozone is formed mainly by

- a) UV radiation splitting O_2 molecules in the stratosphere
- b) Reactions between NO_x and volatile organic compounds (VOCs) in the presence of sunlight
- c) Direct emission from the ocean surface
- d) Reaction of CO_2 with water vapour

Ans: _____

32. The long-term temperature goal of the Paris Agreement, as stated in the slides, is to keep the global temperature rise

- a) Well below 2°C , and preferably limited to 1.5°C , above pre-industrial levels
- b) Below 5°C above pre-industrial levels
- c) At exactly 1°C above pre-industrial levels
- d) Unspecified, with no numerical target

Ans: _____

33. Compared with coal, electricity generation from efficient natural gas plants emits roughly how much less CO_2 per unit of electrical energy?

- a) 10%
- b) 30%
- c) 60%
- d) 90%

Ans: _____

34. Direct Air Capture and Storage (DACS) differs from Carbon Capture and Storage (CCS) mainly in that DACS

- a) Works only on CO_2 from vehicle exhaust
- b) Removes CO_2 directly from ambient air rather than from point sources such as power plants
- c) Cannot be combined with storage
- d) Uses only solid sorbents, never liquid solvents

Ans: _____

35. In solvent-based direct air capture, CO_2 is first reacted with a strong basic hydroxide solution such as

- a) NaOH or KOH
- b) HCl
- c) H_2SO_4
- d) NH_3

Ans: _____

36. Regenerating CO_2 from the CaCO_3 formed during solvent-based DAC (decomposing it back to CO_2 and CaO) requires temperatures of about

- a) $50\text{--}120^\circ\text{C}$
- b) $200\text{--}300^\circ\text{C}$
- c) $\sim 900^\circ\text{C}$
- d) Room temperature

Ans: _____

37. By IUPAC classification, pores smaller than 2 nm in a porous solid sorbent such as a zeolite or MOF are called

- a) Macropores
- b) Mesopores
- c) Micropores

d) Nanopores

Ans: _____

38. Which statement correctly distinguishes chemisorption from physisorption?

- a) Chemisorption involves weak van der Waals forces; physisorption involves strong chemical bonding
- b) Chemisorption is surface-specific with stronger bonding ($\Delta H_{\text{ads}} \approx 50\text{--}500\text{ kJ/mol}$) than physisorption ($\Delta H_{\text{ads}} \approx 5\text{--}50\text{ kJ/mol}$)
- c) Physisorption always requires higher temperatures than chemisorption
- d) Physisorption is always surface-specific, unlike chemisorption

Ans: _____

39. The 2025 Nobel Prize in Chemistry, mentioned in the slides, was awarded for the development of

- a) CRISPR gene editing
- b) Metal–organic frameworks (MOFs)
- c) mRNA vaccine technology
- d) Lithium-ion batteries

Ans: _____

40. Of the roughly 130 Mt of CO_2 used annually to manufacture urea, salicylic acid, and polycarbonates, which process consumes the largest share of CO_2 industrially?

- a) Salicylic acid synthesis
- b) Polycarbonate synthesis
- c) Urea synthesis
- d) Methanol synthesis

Ans: _____

41. Salicylic acid is industrially produced from phenol and CO_2 via which named process?

- a) Haber–Bosch process
- b) Ostwald process
- c) Kolbe–Schmitt process
- d) Fischer–Tropsch process

Ans: _____

42. Bioremediation of environmental pollutants relies mainly on

- a) High-temperature incineration
- b) Living organisms (bacteria, fungi, algae) that convert pollutants into CO_2 , H_2O , and biomass
- c) Chemical precipitation using heavy metals
- d) UV photolysis alone

Ans: _____

43. Which oil-spill remediation method uses physical tools such as booms, skimmers, and vacuum trucks without changing the oil's chemical composition?

- a) In-situ burning
- b) Mechanical remediation
- c) Chemical remediation
- d) Biological remediation

Ans: _____

44. In the microbial degradation of fatty acids and hydrocarbons, the pathway in which oxidation occurs at the C-3 (β) position is called

- a) Glycolysis
- b) The citric acid (TCA) cycle
- c) β -oxidation
- d) The electron transport chain

Ans: _____

45. During microbial degradation of detergents and petroleum, molecular oxygen used for terminal oxidation of the hydrocarbon chain is activated by which enzyme?

- a) Cytochrome P-450
- b) Catalase
- c) Lipase
- d) Amylase

Ans: _____

46. Catalytic oxidation of benzene with air (2–5 bar, 400–450 °C) primarily produces

- a) Phenol
- b) Maleic anhydride
- c) Phthalic acid
- d) Nitrobenzene

Ans: _____

47. The 12 Principles of Green Chemistry were formulated in 1998 by

- a) Fritz Haber and Carl Bosch
- b) Paul Anastas and John Warner
- c) Susumu Kitagawa and Omar Yaghi
- d) Richard Eisenberg and Harry Gray

Ans: _____

48. According to the waste-management hierarchy in the Pollution Prevention Act (PPA) described in the slides, the correct order of preference is

- a) Prevent/reduce at source → recycle → treat → dispose as a last resort
- b) Dispose → treat → recycle → prevent
- c) Recycle → prevent → treat → dispose
- d) Treat → dispose → prevent → recycle

Ans: _____

49. On a GHS chemical hazard label, which pictogram (black symbol on white background, red diamond frame) specifically denotes acute toxicity that may be fatal?

- a) Exclamation mark
- b) Skull and crossbones
- c) Flame over circle
- d) Corrosion

Ans: _____

50. In green chemistry metrics, percentage yield is calculated as

- a) Theoretical yield / Actual yield × 100
- b) Actual yield / Theoretical yield × 100
- c) Actual yield × Theoretical yield
- d) (Actual yield – Theoretical yield) × 100

Ans: _____

Answer Key

1.	c	26.	c
2.	c	27.	b
3.	a	28.	a
4.	c	29.	c
5.	a	30.	b
6.	b	31.	b
7.	b	32.	a
8.	b	33.	c
9.	c	34.	b
10.	c	35.	a
11.	a	36.	c
12.	c	37.	c
13.	b	38.	b
14.	c	39.	b
15.	a	40.	c
16.	b	41.	c
17.	a	42.	b
18.	b	43.	b
19.	b	44.	c
20.	a	45.	a
21.	c	46.	b
22.	b	47.	b
23.	a	48.	a
24.	a	49.	b
25.	b	50.	b